Growth of Functions Recap

$f(x)$ is $\mathcal{O}(g(x))$ if $\exists k$ and $\exists C$ (k and c are called witnesses), such that
$|f(x)| \leq C \times |g(x)|$ whenever $x \geq k$

Example

Suppose $f(x)$ is $5000x^7$ and
$g(x)$ is $0.02x^8$

We can show $f(x)$ is $\mathcal{O}(g(x))$ but $g(x)$ is not $\mathcal{O}(f(x))$

Takeaway, if f and g represent runtime, always pick f in a race over a large problem.

Rough Working:
What condition do we need on x and C to conclude

$|5000x^7| \leq C|0.02x^8|$

1. Divide both sides by x^7 (Wont swap inequality if $x^7 > 0$)

$5000 \leq C0.02x$
or
$\frac{250000}{C} \leq x$ (We can pick any posotive C as a witness and then pick $k = \frac{250000}{c}$)

Note: Bigger C, smaller k

If $C = 250000$ $k = 1$ works

If $C = 50000$, we need $k = 5$

So $f(x)$ is $\mathcal{O}(g(x))$

Why cant we have $g(x)$ being $\mathcal{O}(f(x))$?

What condition do we need on x to make sure

$|0.02x^8| \leq C\ |5000x^7|$ (This is always eventually false)

equally we have $x \leq 250000C$

Example

If $f(x)$ is x^3 and

$g(x)$ is $(1.5 + \sin(x))x^3$

Sometimes, $f(x) > g(x)$,
Sometimes $g(x) > f(x)$

But always
$0.5 \leq (1.5 \sin(x)) \leq 2.5$ so

$|f(x)| \leq 2|g(x)|$ and

$|g(x)| \leq \frac{5}{2}\ |f(x)|$

$|g(x)| = (1.5 + \sin(x))f(x) \leq 2.5f(x)$

$|g(x)| = (1.5 + \sin x)f(x) \geq 0.5f(x)$

So $f(x)$ is $\mathcal{O}(f(x))$ with witnesses $C = \frac{5}{2}, k = 1$

Theorem 1

If $f(x) = a_n x^{n-1} + a_1 x + a_0$

then

$f(x)$ is $\mathcal{O}(x^n)$ with witnesses $k = 1$ and $C = |a_n| + |a_{n-1}| + ... + |a_1| + |a_0|$

We can get a smaller C with a bigger k

A convinient list of functions comes after example 7 in the text.

$1, \log n, n, n \log n, n^2, 2^n, n!$ (These are all non decreasing functions) (Growing to infinity except 1)

Each of these functions is $\mathcal{O}$ of every function to its right. (Also not $\mathcal{O}$ of any function to its left)

Example

$n \log n$ is $\mathcal{O}(n^2)$

n is $\mathcal{O}(2^n)$

$n \log n$ is not $\mathcal{O}(n)$

The math is calculus we could show

$\lim_{n \to \infty} \frac{f(x)}{g(x)} = 0$ whenever f is left of g

Calculus then tells us an appropriate k exists, but not how to find it.

Note

This limit interpretation did not apply to x^3 vs $(1.5 + \sin x)x^3$

$\lim_{n \to \infty} \frac{n^3}{(1.5 + \sin n)n^3}$ does not exist.

Some Gaps

Starting point: $1, \log n, n, n \log n, n^2, 2^n, n!$

If $d > c > 1$ then

n^c is $\mathcal{O}\left(n^d\right)$ but not the reverse

$C = 1$ and $d = 2$ are on the list.

If $b > 1$ and cd are positive, then

$(\log_b n)^c$ is $\mathcal{O}\left(n^d\right)$

Example

$(\log_{10} n)^{5000}$ is $\mathcal{O}\left(n^{0.0000000125}\right)$

- 5000 is C

n^d is $\mathcal{O}(b^n)$ (as long as $b > 1$ and d is positive)

Example

$2650x^{782}$ is $\mathcal{O}(1.003^x)$

Combining Functions

Theorem

If $f_1(x)$ is $\mathcal{O}(g_1(x))$ and

$f_2(x)$ is $\mathcal{O}(g_2(x))$ then

$(f_1 + f_2)(x)$ is $\mathcal{O}(\max(|g_1(x)|, |g_2(x)|))$

Proof Idea

Both hypothesis have witnesses,

k_1, C_1, k_2, C_2

The witness for a conclusion is

$k = \max(k_1, k_2))$

$C = C_1 + C_2$

If $x > k$ then $x > k_1$ and $x > k_2$ so

$|(f_1 + f_2)(x)| \leq |f_1(x)| + |f_2(x)|$

$\leq C_1 \ |g_1(x)| + C_2|g_2(x)|$

$\leq C_1 \max(|g_1(x), |g_2(x)|) + C_2 \max(|g_1(x)|, |g_2(x)|)$

$= (C_1 + C_2)|\max(|g_1(x)|, |g_2(x)|)|$

Application

A two-step algorithm's performance is summarized by the worst behaved stop,

Coronary If $f_1(x)$ is $\mathcal{O}(g(x))$ and $f_2(x)$ is $\mathcal{O}(g(x))$ then

$(f_1 + f_2)(x)$ is $\mathcal{O}(g(x))$

Combining Functions

Theorem:

If $f_1(x)$ is $\mathcal{O}(g_1(x))$ and

$f_2(x)$ is $\mathcal{O}(g_2(x))$ then

$(f_1 f_2)(x)$ is $\mathcal{O}(g_1(x)g_2(x))$

Complementary Notation

Definition

$f(x)$ is $\Omega(g(x))$ if $\exists k, \exists C$ so that $x > k$ implies

$|f(x)| \geq C\ |g(x)|$

compare $f(x)$ is $\Omega(g(x))$ to $g(x)$ is $\mathcal{O}(f(x))$

Definition

If $f(x)$ is $\mathcal{O}(g(x))$ and

$f(x)$ is $\Omega(g(x))$ then,

$f(x)$ is $\Theta(g(x))$ (Big Theta)

In this case we say $f(x)$ and $g(x)$ have the same order,

Example

If $f(x) = x^3$ and

$g(x) = (1.5 + \sin x)x^3$ then

$f(x)$ is $\Theta(g(x))$

When two functions are the same order, the winner is the one on the faster computer.

Theorem

If $f(x) = a_n x^n + a_{n-1}x^{n-1} + ... + a_1 x + a_0$

and $a_n \neq 0$

then

$f(x)$ is $\Theta(x^n)$